Digvijay Singh

Permanent Resident of the USA | digvijay.in1@gmail.com | www.digvijay.org | | PubMed.gov | | | | |

Summary

Last Updated on 9/2026

- Interdisciplinary scientist spanning experimental biology, assay/platform development, quantitative analysis, and computational/AI methods.
- 23 publications (9 first/co-first; 2X Cell, Nature family, etc.; ¿3000 citations); NIH K99-R00 & Damon Runyon awardee.

EDUCATION

- **Ph.D. - Johns Hopkins University School of Medicine** 8/2012 - 1/2018
- **Integrated BS-MS - Indian Institute of Technology (IIT), Kharagpur** 7/2007 - 4/2012
- **MicroMBA - Rady School of Management, University of California San Diego**

PROFESSIONAL EXPERIENCES

Senior Staff Scientist at Sarmal 10/2026 –
- Developing a new diagnostic platform.

Staff Scientist at Singular Genomics 9/2024 – 7/2026
- Led G4x pathology-assay development from R&D toward production across assay, manufacturing, image-processing, software, optics, and product teams.
- Shipped quantitative/AI tools; improved cell-segmentation PQ from roughly 0.5 to 0.75 and image-registration speed and quality by approximately 35%.
- Identified a surface-adsorption problem; analysis tools I developed helped guide a surface-deposition process transferred into production.

NIH K99-R00 Fellow at University of California (UC), San Diego 2/2023 – 9/2024
- Co-led determination of the Nuclear Basket architecture (Co-first in Cell); helped develop 3 AI-driven biological tools.

Damon Runyon Fellow at UC San Diego 2/2019 – 2/2023
- Determined nuclear-pore structures (Co-first in Cell), studied condensates, & co-authored a book on tomography.

Postdoctoral & Visible Molecular Cell Fellow at UC San Diego 2/2018 – 2/2019
- Established a reproducible workflow for a difficult in-cell structural-biology project, resolving key bottlenecks and driving a highly cited publication.

PhD researcher at Johns Hopkins University & UIUC 6/2012 – 1/2018
- Developed quantitative CRISPR assays defining Cas9/Cas12a target recognition, specificity, kinetics, and product release; produced multiple first/co-first-author papers.

Multiple internships at top-tier universities & companies (awarded with stipends)
Locations: University of Cambridge, Massachusetts Institute of Technology, IIT, KIT-Germany, Univ. of Illinois (UC), Unilever, and General Electric. **Scope:** Computational modeling, protein purification, kinetic analysis, & organic synthesis.

SKILLS

- **Experimental Biology & Assays**: Cell/molecular biology, CRISPR, protein/nucleic-acid biochemistry, cell culture, PCR/cloning, assay optimization/QC/troubleshooting.
- **Quantitative & Computational Biology**: Python, R, Bash; statistics; NGS/multi-omics; scRNA-seq; image analysis; AI/ML.
- **Imaging & Structural Biology**: Fluorescence/smFRET, super-resolution microscopy, cryo-EM/cryo-ET, cryo-FIB, tomography, structural modeling.
- **Product & Scientific Leadership**: R&D toward production; mentoring/training; technical documentation/presentations; cross-functional execution.

SELECT PRESENTATIONS (Invited or Selected): 3DEM GRC (2023), Biophysical Society Meeting (2023, 2022, 2017, 2015), Southern California cryo-EM symposium (2022), Cryo-EM workshop at Brookhaven National Lab, USA (2022, 2023), National Tomography Workshop (3 lectures & workshops each year), AIIMS, New Delhi (2022-2026), Friends of Cell Meeting, San Diego Cluster of Institutes (2022), CRISPR Workshop at CSIR-IGIB, New Delhi (2019), Physics of Living Systems Conference (2015), Center for Physics of Living Cells Symposium, UIUC (2015).

EXPERT SERVICES, TEACHING & MENTORSHIP

- **Reviewer for:** *Nature Structural & Molecular Biology, Proceedings of the National Academy of Science, Nature Communications, Journal of Biological Chemistry, Chemical Science, Biochemistry, ACS Omega, Scientific Reports, Cellular & Molecular Life Sciences, Frontiers in Molecular Neuroscience, Frontiers in Neuroscience, JoVE.*
- **Instructor/Teaching assistant** at *Physics of Living Cells summer schools* & University of Illinois (2013-15).
- **Mentor for** 3 students (UC San Diego), 3 students (Johns Hopkins University)

PUBLICATIONS [23 so far; 9 first/co-first*] [Citations > 3000]

Selected:

19. **D. Singh***, N. Soni*, J. Hutchings*, .., E. Villa. The Molecular Architecture of the Nuclear Basket. *Cell* (2024).
16. C. Akey*, **D. Singh***, C. Ouch*, I. Echeverria*, ... E. Villa, M. Rout. Comprehensive structure & functional adaptations of the yeast nuclear pore complex. *Cell* (2022). Media-Release
4. **D. Singh**, ..., J. A. Doudna, T. Ha. Real-time observation of DNA recognition & rejection by the RNA-guided endonuclease Cas9. *Nature Communications* (2016).
8. **D. Singh**, ..., T. Ha. Real-time observation of DNA target interrogation & product release by .. Cas12a. *PNAS* (2018).
7. **D. Singh**, ..., T. Ha. Mechanism of improved specificity of engineered Cas9s revealed by single molecule analysis. *Nature Structural & Molecular Biology* (2018).
15. L. Dai*, **D. Singh***, ..., V. B. Rao. A viral genome packaging ring-ATPase is a flexibly coordinated pentamer. *Nature Communications* (2021).
9. I. Okafor*, **D. Singh***, Y. Wang*, .., T. Ha. Single molecule analysis of effects of non-canonical guide RNAs & specificity-enhancing mutations on Cas9-induced DNA unwinding. *Nucleic Acids Research* (2019).
18. **D. Singh**, E. Villa. Cryo-Focused Ion Beam Milling of Cells. *Springer book on Cryo-Electron Tomography* (2023).
6. **D. Singh**, T. Ha. Understanding the molecular mechanism of CRISPR toolbox using single-molecule approaches. *ACS Chemical Biology* (2018).
3. J. Fei, **D. Singh**,... , T. Ha. Determination of in vivo target search kinetics of regulatory noncoding RNA. *Science* (2015).

Remaining:

13. H. Yu, .., **D. Singh**, ..HSP70 chaperones RNA-free TDP-43 into anisotropic intranuclear liquid spherical shells. *Science* (2020).
12. S. Lu, .., **D. Singh**, ..The SARS-CoV-2 Nucleocapsid phosphoprotein forms ...condensates with RNA & ..the M protein. *Nature Communi.* (2020).
20. Y. Liang, .., **D. Singh**, ..RobPicker: A Meta Learning Framework for Robust Identification of Macromolecules in Cryo-Electron Tomograms.
21. Y. Liang, ..., **D. Singh**, ..Multi-Modal Large Language Model Enables All-Purpose Prediction of Drug Mechanisms & Properties.
22. Li Zhang, .., **D. Singh**, ..A tri-modal contrastive learning framework for protein representation learning. *Cell Reports* (2026).
11. Y. Wang, .., **D. Singh**, ..Real-time observation of Cas9 postcatalytic domain motions. *PNAS* (2020).
10. F. Wagner, .., **D. Singh**, ..Preparing samples from whole cells using focused-ion-beam milling for cryo-electron tomography. *Nature Protocols* (2020).
1. A. Dhar, .., **D. Singh**,..Protein stability & folding kinetics in nucleus & endoplasmic reticulum of eukaryotic cells. *Biophysical Journal* (2011).
2. B. Hua, .., **D. Singh**, ..An improved surface passivation method for single-molecule Studies. *Nature Methods* (2014).
17. B. Raveh, .., **D. Singh**, ..Integrative mapping reveals molecular features underlying the mechanism of nucleocytoplasmic transport. *PNAS* (2025).
23. A. Abadia, .., **D. Singh**, ..TissuStamp: A high-throughput tissue transfer workflow for FFPE spatial biology . *Journal of Histotechnology* (2026).
5. B. Hua, .., **D. Singh**, ..Single-molecule centroid localization algorithm improves the accuracy of fluorescence binding assays. *Biochemistry* (2018).
14. A. Poddar, .., **D. Singh**, ..Effects of individual base-pairs on in vivo target search & destruction kinetics of small RNA. *Nature Communi.* (2021).

.. → *Please refer to the link for complete author list.*

AWARDS & FUNDING

- NIH K99-R00 Pathway to Independence Award, National Institute of Health. 2023 - 2028
- Damon Runyon Fellowship, Damon Runyon Cancer Foundation 2019 - 2023
- Virtual Molecular Cell Consortium (VMCC) Fellow, UC San Diego. 2018 - 2019
- Finalist of Damon Runyon-Dale F. Frey Award for Breakthrough Scientists 2022
- Biophysical Society Education Travel Award 2017
- Finalist of International Howard Hughes Medical Institute fellowship 2015
- Johns Hopkins Biophysics department nominee for international Weintraub Award 2018
- INSPIRE fellowship by Government of India 2008 - 2012